Business Sales Performance Analysis

Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Row ID	Text	Unique row identifier
Customer ID	Text	Unique ID for each customer
Product ID	General	Unique product code
Order ID	Text	Unique ID for each order
Customer Name	Text	Name of the customer
Segment	Text	Customer group/category
Country	Text	Country of the customer
City	Text	City of the customer
State	Text	State of the customer
Postal Code	Text	Area zip code
Region	Text	Location of sale
Category	Text	Group of products
Sub-Category	Text	Product sub-group
Product Name	Text	Name of the product
Order Date	Date	Date of purchase
Ship Date	Date	Date of shipment
Sales	Currency	Revenue from the order
Discount	Percentage	Reduction applied on price
Profit	Currency	Earnings after costs

Tools & Technologies

Excel: *Data cleaning, transformation, Sorting, Created separate tables using filters*

Power BI: *Data modeling, DAX measures & Visualization*

Tasks Performed:

Data Cleaning & Transformation (Excel):

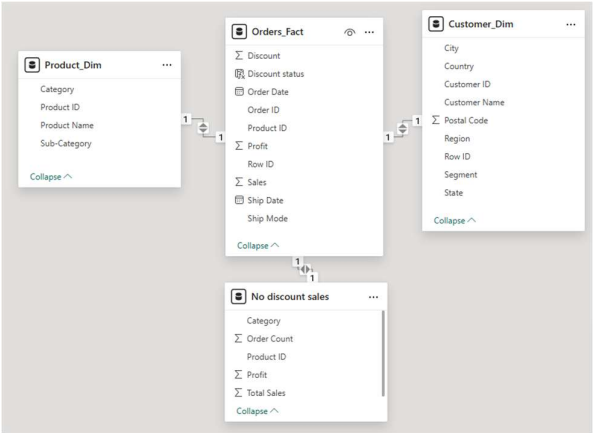
- *Data cleaning and transformation were performed in Excel to ensure data quality and consistency.*
- *Trimming and cleaning techniques were applied to remove unnecessary spaces and errors.*
- *The dataset was structured by splitting it into three tables: Customer_Dim, Product_Dim, and Orders_Fact to support efficient data modeling.*
- *Product names were standardized into shorter and more readable formats.*
- *Date formats were corrected for consistency, and Product ID was refined as a unique identifier to eliminate duplicates.*
- *Additionally, overall duplicate records were removed to maintain data accuracy.*

Data Modeling & Visualization (PowerBI):

- *Data modeling was implemented in Power BI by establishing relationships between the tables.*
- *A new column, Discount Status, was created in the Orders_Fact table to classify transactions as "No Discount" or "Discount Applied."*
- *Multiple analytical tables were generated, including city-wise, state-wise, region-wise, segment-wise, and no-discount sales to enable detailed analysis.*
- *DAX measures were created to calculate key metrics, and interactive dashboards were developed using cards, charts, and slicers to provide dynamic and insightful visualizations.*

Data Modelling and DAX (Power BI)

Data Model:



Calculated Columns & DAX Measures:

Caculated table: DAX formula

City-wise sales =

```
SUMMARIZE(Customer_Dim,Customer_Dim[City],"Total Sales",calculate(sum(Orders_Fact[Sales])))
```

City	Total Sales
Los Angeles	$1,75,851.33
San Francisco	$1,12,669.09
San Diego	$47,521.05
Roseville	$3,588.63
Pasadena	$6,590.16
San Jose	$6,071.22
Redlands	$2,891.02
Whittier	$444.77
Santa Clara	$222.19
Brentwood	$5,090.12
Inglewood	$3,479.33
Long Beach	$13,455.02
Hesperia	$1,213.17
Huntington Beach	$2,527.82
Concord	$10,542.4

Region-wise sales =

```
SUMMARIZE(Customer_Dim,Customer_Dim[Region],"Total Sales", CALCULATE(SUM(Orders_Fact[Sales])))
```

Region	Total Sales
West	$7,25,457.93
Central	$5,01,239.88
East	$6,78,781.36
South	$3,91,721.9

State-wise sales =

```
summarize(Customer_Dim,Customer_Dim[State],"Total sales",calculate(sum(Orders_Fact[Sales])))
```

State	Total sales
California	$4,57,687.68
Texas	$1,70,187.98
Illinois	$80,166.16
Oregon	$17,431.14
Ohio	$78,258.21
Arizona	$35,282.02
Louisiana	$9,217.03
New Jersey	$35,764.32
Florida	$89,473.73
Minnesota	$29,863.15

No discount sales =

```
SUMMARIZE(filter(Orders_Fact,Orders_Fact[Discount]=0),Orders_Fact[Product
ID],Product_Dim[Category],Orders_Fact[Profit],"Total Sales",sum(Orders_Fact[Sales]),"Order
Count",count(Orders_Fact[Order ID]))
```

Product ID	Category	Profit	Total Sales	Order Count
07-OFF-NEWELL	Office Supplies	1.9656	7.28	1
010-OFF-BELKIN	Office Supplies	34.47	114.9	1
017-OFF-STURDSTOR	Office Supplies	13.3176	665.88	1
022-OFF-NEWELL	Office Supplies	5.0596	19.46	1
023-OFF-SIXOUTLET	Office Supplies	15.6884	60.34	1
043-OFF-ELDON	Office Supplies	3.894	77.88	1

Segment-wise Sales =

```
SUMMARIZE(Customer_Dim,Customer_Dim[Segment],"Total Sales",calculate(sum(Orders_Fact[Sales])))
```

Segment	Total Sales
Consumer	$11,61,401.34
Corporate	$7,06,146.44
Home Office	$4,29,653.29

Caculated Column: DAX formula

Discount status =

```
if(Orders_Fact[Discount]=0,"No Discount","Discount Applied")
```

Discount	Profit	Discount status
0%	41.91	No Discount
0%	219.58	No Discount
0%	6.87	No Discount
45%	-383.03	Discount Applied
20%	2.52	Discount Applied
0%	14.17	No Discount
0%	1.97	No Discount
20%	90.72	Discount Applied

Calculated Measures: DAX formula

Customer Count = `DISTINCTCOUNT(Customer_Dim[Customer Name])`

Overall Profit = `sum(Orders_Fact[Profit])`

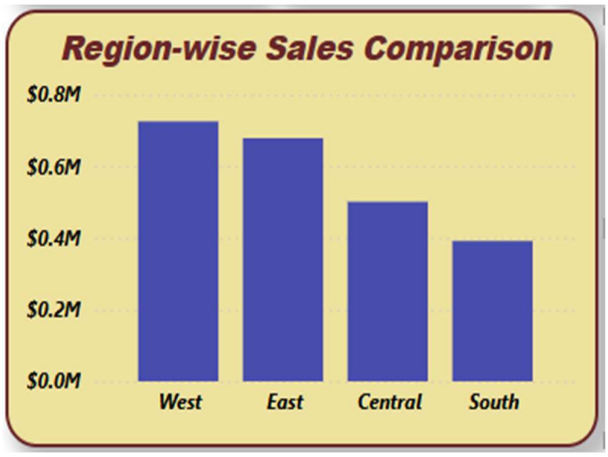
Total Orders = `count(Customer_Dim[Row ID])`

Total Revenue = `sum(Orders_Fact[Sales])`

Visualizations (Power BI)

Mutliple Visualizations:

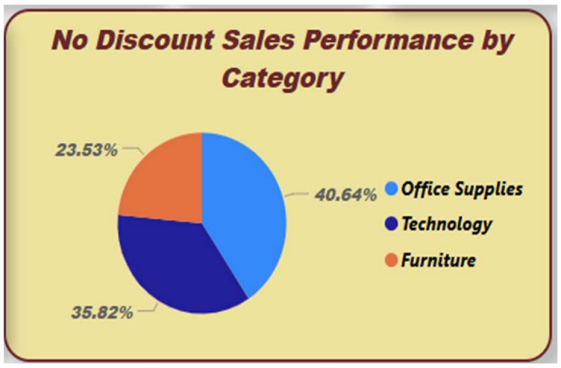
- ***Stacked Column Chart:***



Insights:

- High sales does not always mean high profit
- Discounts can negatively impact profitability
- Some categories underperform despite high demand
- Sales trends and profit trends are not always aligned
- No-discount sales show true business performance

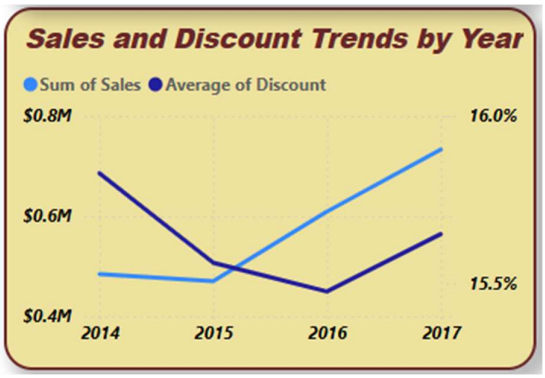
- ***Pie Chart:***



Insights:

- Office Supplies leads in no-discount sales
- Technology shows strong independent demand
- Furniture depends more on discounts
- High no-discount sales indicate strong product value

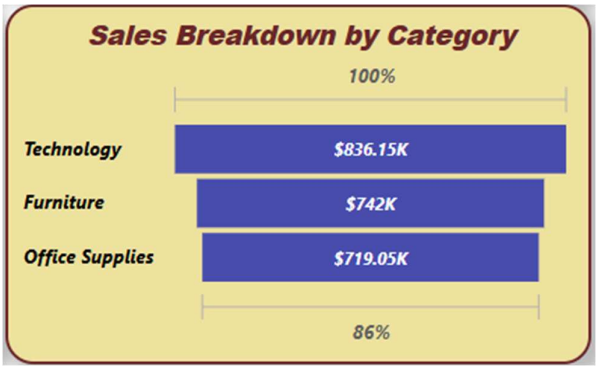
- ***Line Chart:***



Insights:

- Sales stayed flat in 2014–2015, then rose strongly through 2017
- Discounts dropped from 2014 to 2016, then increased in 2017
- Lowest discount point was 2016 (while sales were rising)
- 2017 had both highest sales and a rebound in discounts

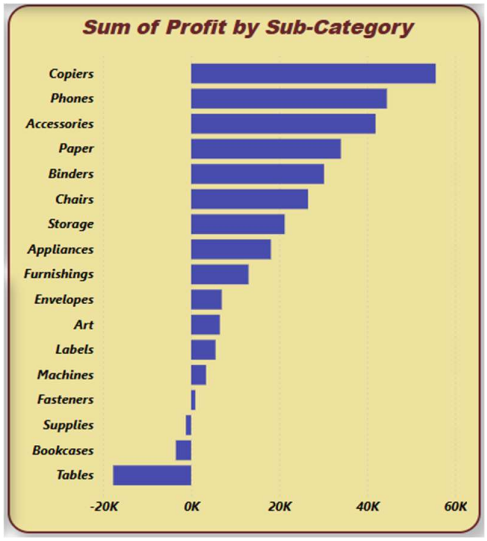
- ***Funnel Chart:***



Insights:

- Technology leads with highest sales
- Furniture is second, slightly behind
- Office Supplies is lowest but close to Furniture
- Small gap overall—categories perform similarly

- ***Clustered Bar Chart:***



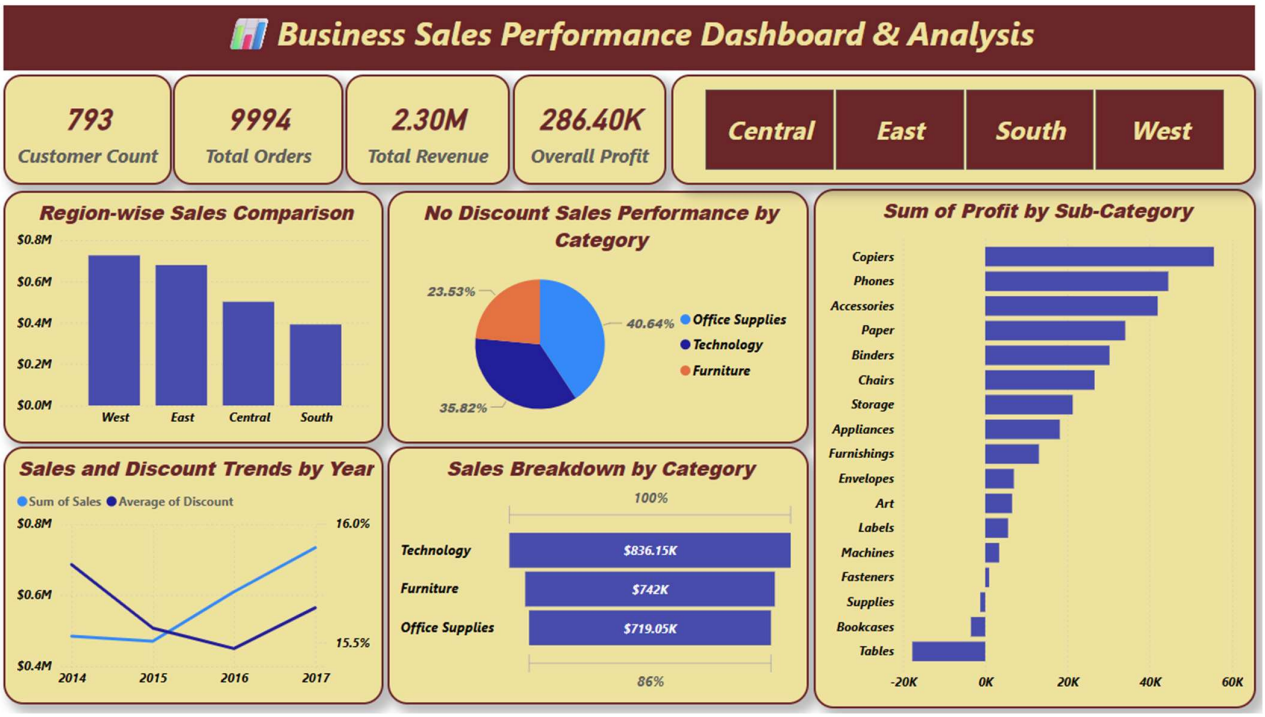
Insights:

- Copiers generate the highest profit
- Phones and Accessories are strong contributors
- Mid-tier: Paper, Binders, Chairs
- Low profit: Envelopes, Art, Labels
- Loss-makers: Tables (largest loss), Bookcases slightly negative
- Most sub-categories are profitable overall

- ***Slicers:***

- ***Cards:***

Dashboards:



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